Diabetes Mellitus

How insulin, blood glucose, and health are connected

Example Student · BIOL 110 · The Community College of Baltimore County

The central idea: Diabetes mellitus is a group of conditions in which blood glucose stays too high because insulin is absent, insufficient, or not working effectively. [1, 2]

Blood glucose regulation

After a meal	Between meals
Blood glucose rises → pancreatic beta cells release **insulin** → tissues take up or store more glucose; the liver reduces glucose release.	Blood glucose falls → the pancreas releases **glucagon** → the liver releases stored glucose.

Diabetes disrupts this regulation through changes in insulin production, insulin action, or both. [1, 2]

Three major types

Type 1 — An autoimmune process damages insulin-producing beta cells. Little or no insulin is made; insulin therapy is essential. [1, 2]

Type 2 — Tissues become less responsive to insulin, and beta cells may not meet the body's needs. Multiple factors contribute. [1, 2]

Gestational — High blood glucose develops during pregnancy. Follow-up testing after birth matters because future type 2 risk is higher. [1, 2]

Possible signs & diagnosis

Possible signs include increased thirst or urination, hunger, fatigue, blurry vision, or unexplained weight loss. Some people have no noticeable symptoms. [2]

Diabetes-range results for nonpregnant people [1, 3]

Test	Result
A1C	**≥ 6.5%**
Fasting plasma glucose	**≥ 126 mg/dL**
2-hour glucose tolerance test	**≥ 200 mg/dL**
Random plasma glucose	**≥ 200 mg/dL** with classic symptoms or hyperglycemic crisis

Without unequivocal hyperglycemia, an abnormal result is generally confirmed with repeat testing. Pregnancy uses different criteria. [1, 3]

Protecting health

Long-term high blood glucose can damage blood vessels and nerves, raising the risk of eye, kidney, nerve, and heart disease or stroke. Complications are not inevitable. [1, 2]

- **Type 1:** insulin is essential.
- **Type 2:** individualized nutrition, activity, medicines, and/or insulin.
- **All types:** monitoring and preventive checks with a health-care team. [4]
- Healthy habits or programs can lower type 2 risk, not eliminate it; type 1 has no established prevention. [2]

Sources · [1] American Diabetes Association Professional Practice Committee. *Diabetes Care*. 2025;48(Suppl. 1):S27–S49. doi:10.2337/dc25-S002. · [2] CDC: About diabetes. · [3] NIDDK: Diabetes tests & diagnosis. · [4] NIDDK: Diabetes treatments.

Educational overview only—not a diagnostic tool or personal treatment advice.